



TOP 5000 MOVIES
DATA ANALYSIS AND
LINEAR REGRESSION



What's playing?



**The
Shawshank
Redemption**



**The
Godfather**



**The Dark
Knight**



**The Lord of
the Rings:
The Return of
the King**

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IMDb Top 250 Movies

As rated by regular IMDb voters.

Ranking



Kalki 2898 AD

7.2/10 Rating



Shaitaan

8/10 Rating



Munjya

9.2/10 Rating



Fighter

8.3/10 Rating

#1 What is IMDb

- IMDb = Internet Movie Database
- Contains ratings from millions of viewers
- Dataset used: Top 5000 movies



Dataset Introduction



a. Title Constants

b. Movie Title

c. Release Year

d. Ranking

e. Overall Rating

f. Number of Voters

g. Runtime(in min)

h. Directors

i. Writers

j. Genres

k. IMDb Link

l. Title_IMDb_Link

	iconId	primaryTitle	startYear	rank	overallRating	numVotes	runtimeMinutes	directors	writers	genres	IMDbLink	Title_IMDb_Link
0	tt0111161	The Shawshank Redemption	1994	1	9.3	3120155	142	Frank Darabont	Stephen King, Frank Darabont	Drama	<a href="https://www.imdb.com/title/tt0111161"...	<a href="https://www.imdb.com/title/tt0111161"...
1	tt0068646	The Godfather	1972	2	9.2	2175958	175	Francis Ford Coppola	Mario Puzo, Francis Ford Coppola	Crime, Drama	<a href="https://www.imdb.com/title/tt0068646"...	<a href="https://www.imdb.com/title/tt0068646"...
2	tt0468569	The Dark Knight	2008	3	9.1	3095865	152	Christopher Nolan	Jonathan Nolan, Christopher Nolan, David S. Goyer	Action, Crime, Drama	<a href="https://www.imdb.com/title/tt0468569"...	<a href="https://www.imdb.com/title/tt0468569"...
3	tt0167260	The Lord of the Rings: The Return of the King	2003	4	9.0	2119280	201	Peter Jackson	J.R.R. Tolkien, Fran Walsh, Philippa Boyens, P...	Adventure, Drama, Fantasy	<a href="https://www.imdb.com/title/tt0167260"...	<a href="https://www.imdb.com/title/tt0167260"...
4	tt0108052	Schindler's List	1993	5	9.0	1555434	195	Steven Spielberg	Thomas Keneally, Steven Zaillian	Biography, Drama, History	<a href="https://www.imdb.com/title/tt0108052"...	<a href="https://www.imdb.com/title/tt0108052"...

PROJECT GOAL

1. UNDERSTAND WHAT FACTORS INFLUENCE IMDB RATINGS
2. EXPLORE RELATIONSHIPS BETWEEN FEATURES (GENRE, VOTES, RUNTIME, YEAR) AND RATING
3. BUILD A SIMPLE REGRESSION MODEL TO PREDICT THE RATING USING THESE FEATURES
4. COMPARE MODELS (LINEAR, RIDGE, LASSO) USING CROSS-VALIDATION



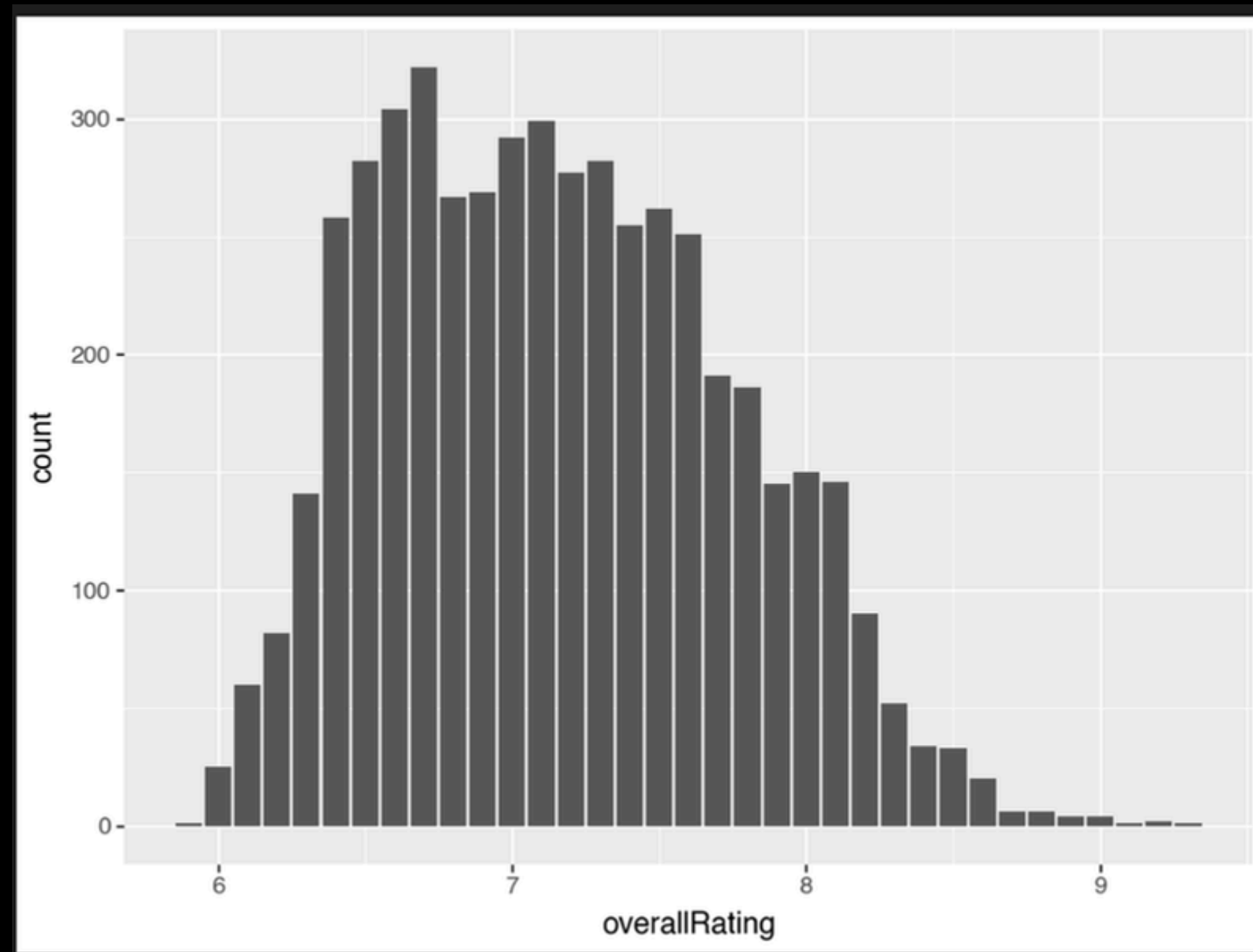
Feature Engineering

Extracting main_genre

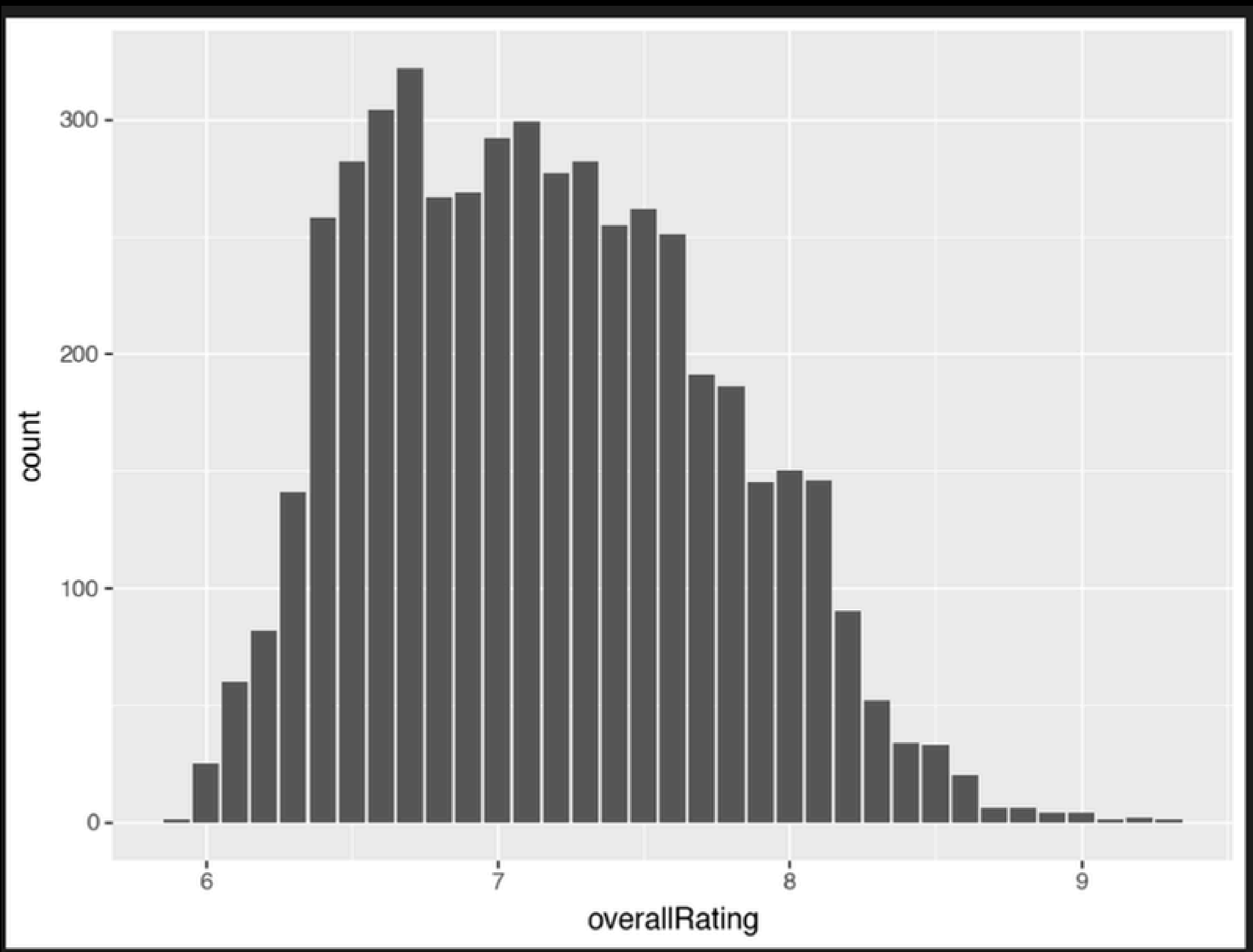
- Movies often have multiple genres (e.g., "Drama, Crime").
- To simplify analysis and improve model performance, the first listed genre is extracted as main_genre.
- This allows us to treat genre as a single categorical variable in regression.

	genres	main_genre
0	Drama	Drama
1	Crime, Drama	Crime
2	Action, Crime, Drama	Action
3	Adventure, Drama, Fantasy	Adventure
4	Biography, Drama, History	Biography

RATING DISTRIBUTION



- Ratings in this dataset range from 5.9 to 9.3
- Most movies cluster between 6.7 and 7.6
- Important context before evaluating model performance



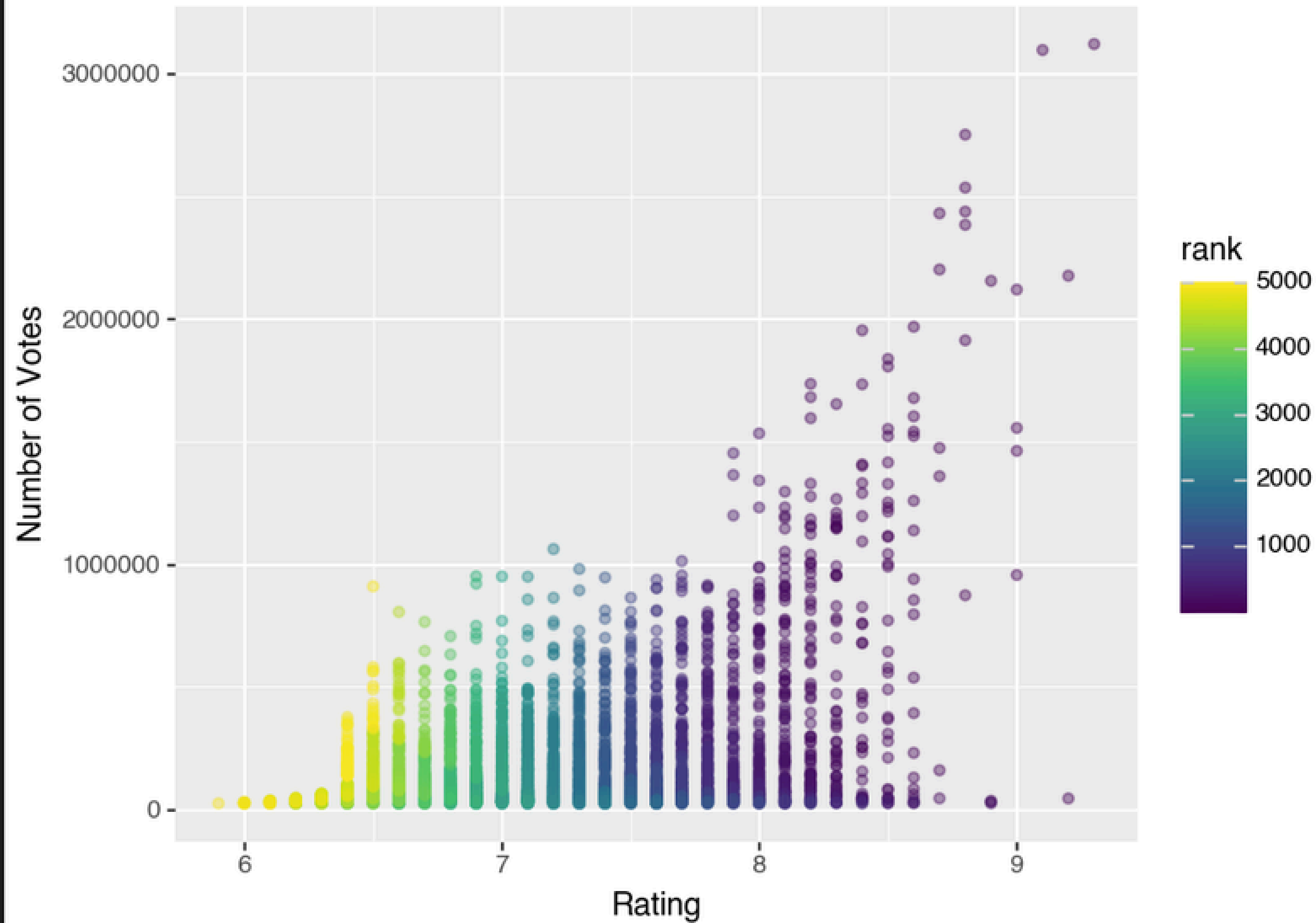
RELATIONSHIP BETWEEN VOTES AND RATING



- Movies with more votes tend to have slightly higher ratings
- A positive trend is visible
- Some outliers exist with high votes but average ratings
- Each point = one movie.
- Movies with more votes tend to have slightly higher ratings.
- Popular movies (high votes) usually reflect stronger audience consensus.
- Outliers exist: some highly-voted movies still have average ratings.



Number of Votes vs Rating



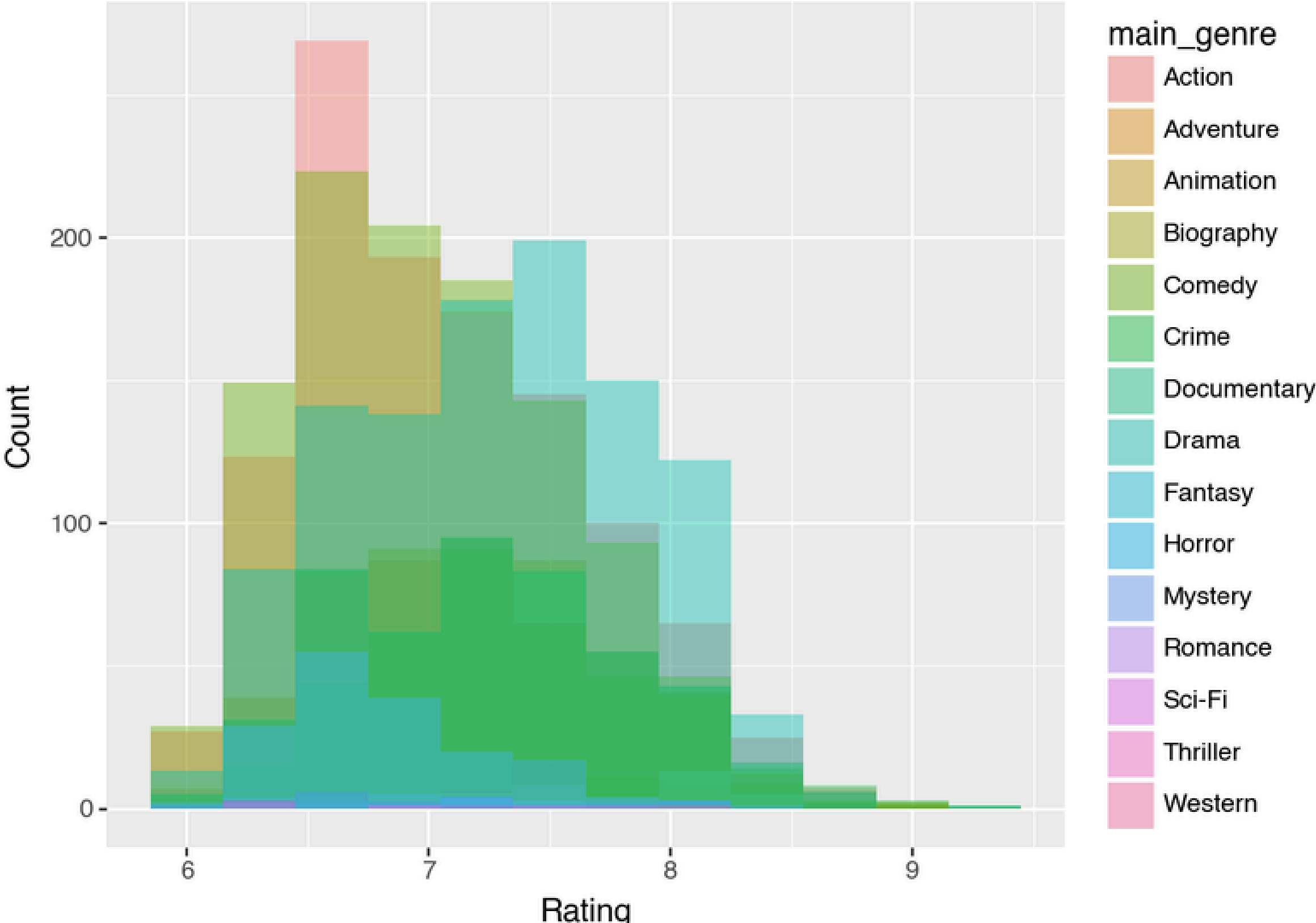
RATING DISTRIBUTION BY GENRE



- Shows how movie ratings are distributed across genres
- Action appears the most because it has the largest number of movies in the dataset
- Some genres (e.g., Western) appear less often but tend to have higher ratings
- Highlights both frequency and spread of ratings across categories



Rating Distribution by Genre



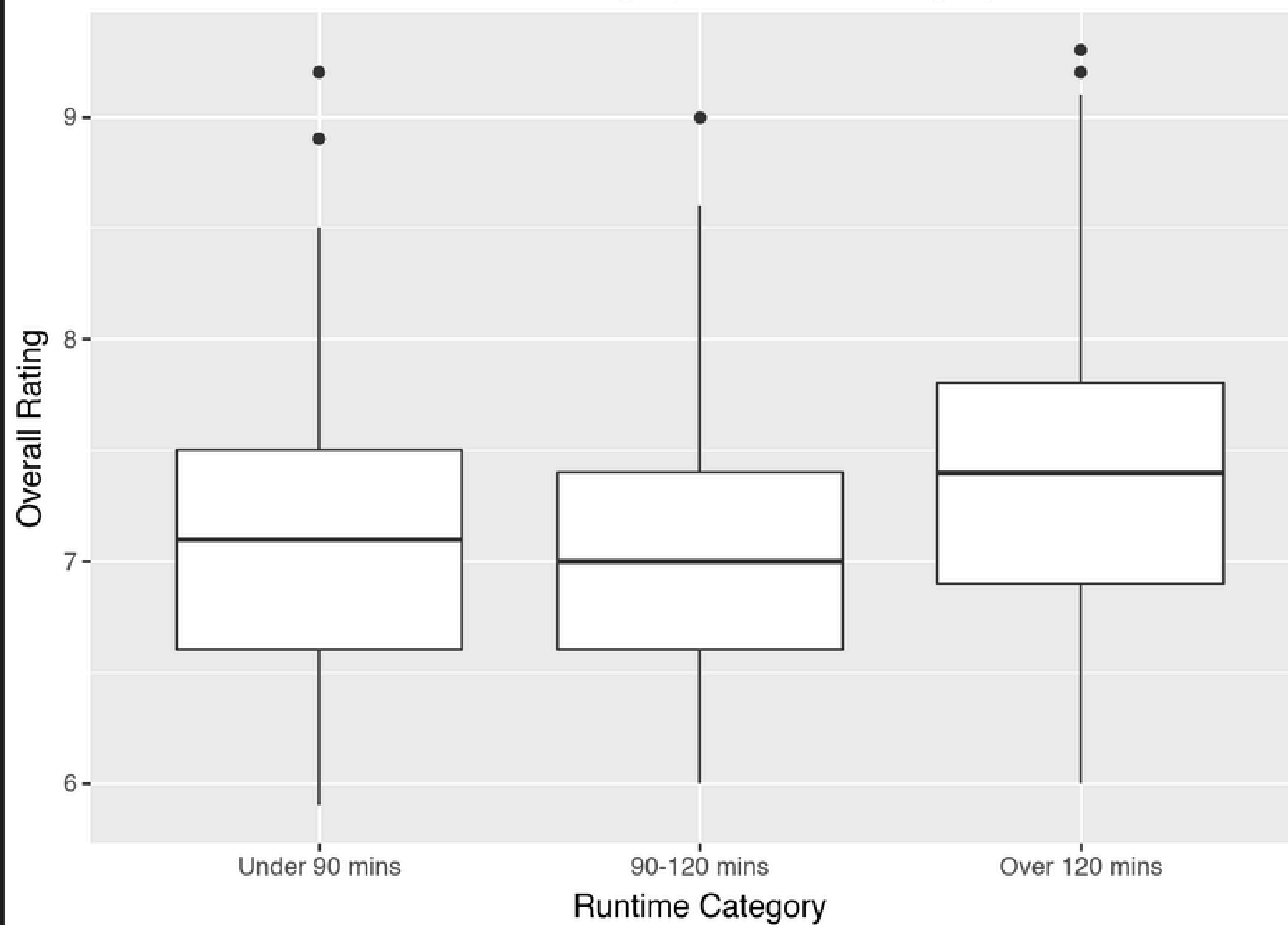
OVERALL RATING BY RUNTIME



- Movies over 120 mins have the highest median rating.
- Under 90 mins and 90–120 mins show similar medians, both lower than long movies.
- Longer movies show wider rating variation.
- Outliers appear in all categories.



Overall Rating by Runtime Category

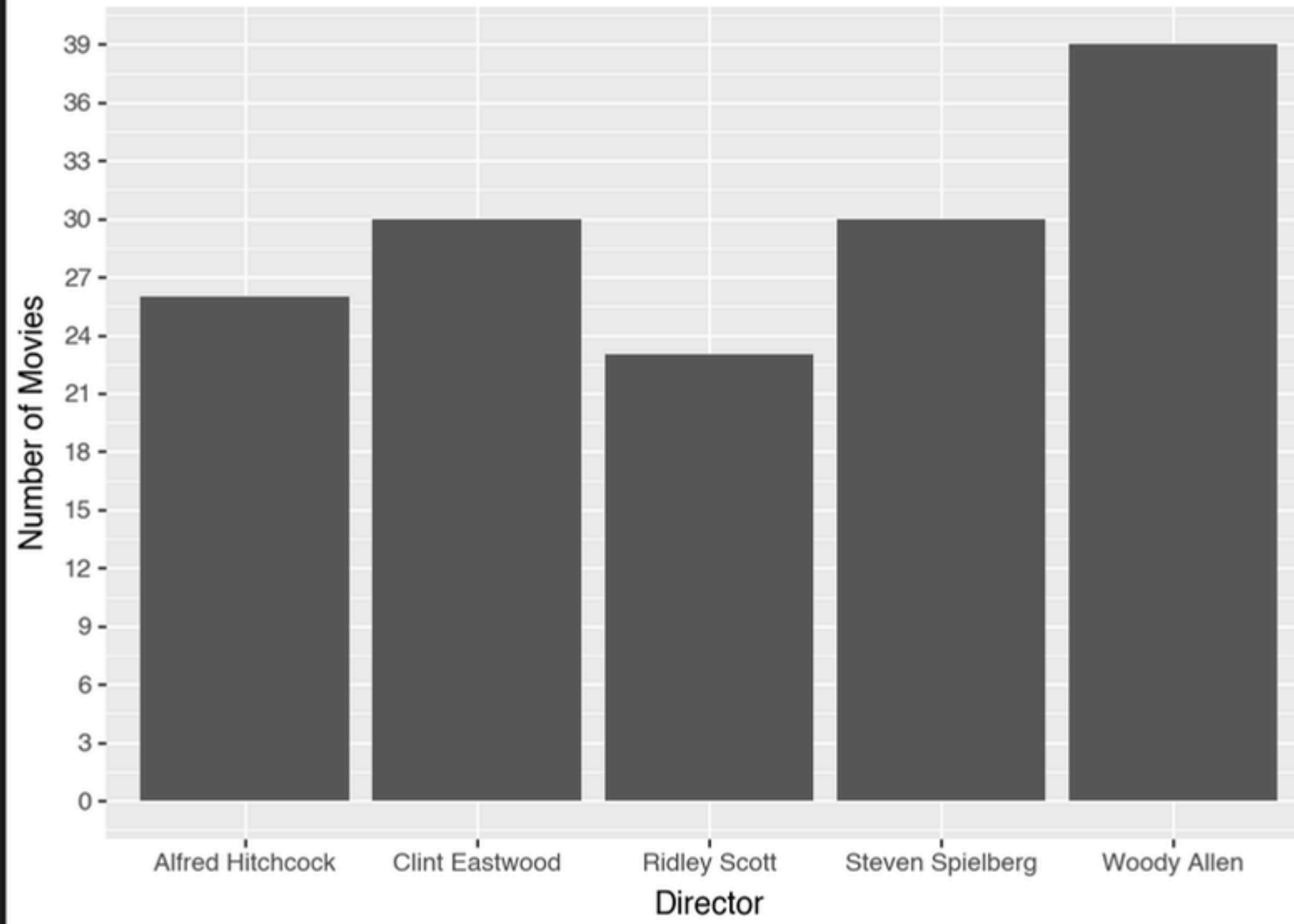


TOP 5 DIRECTORS

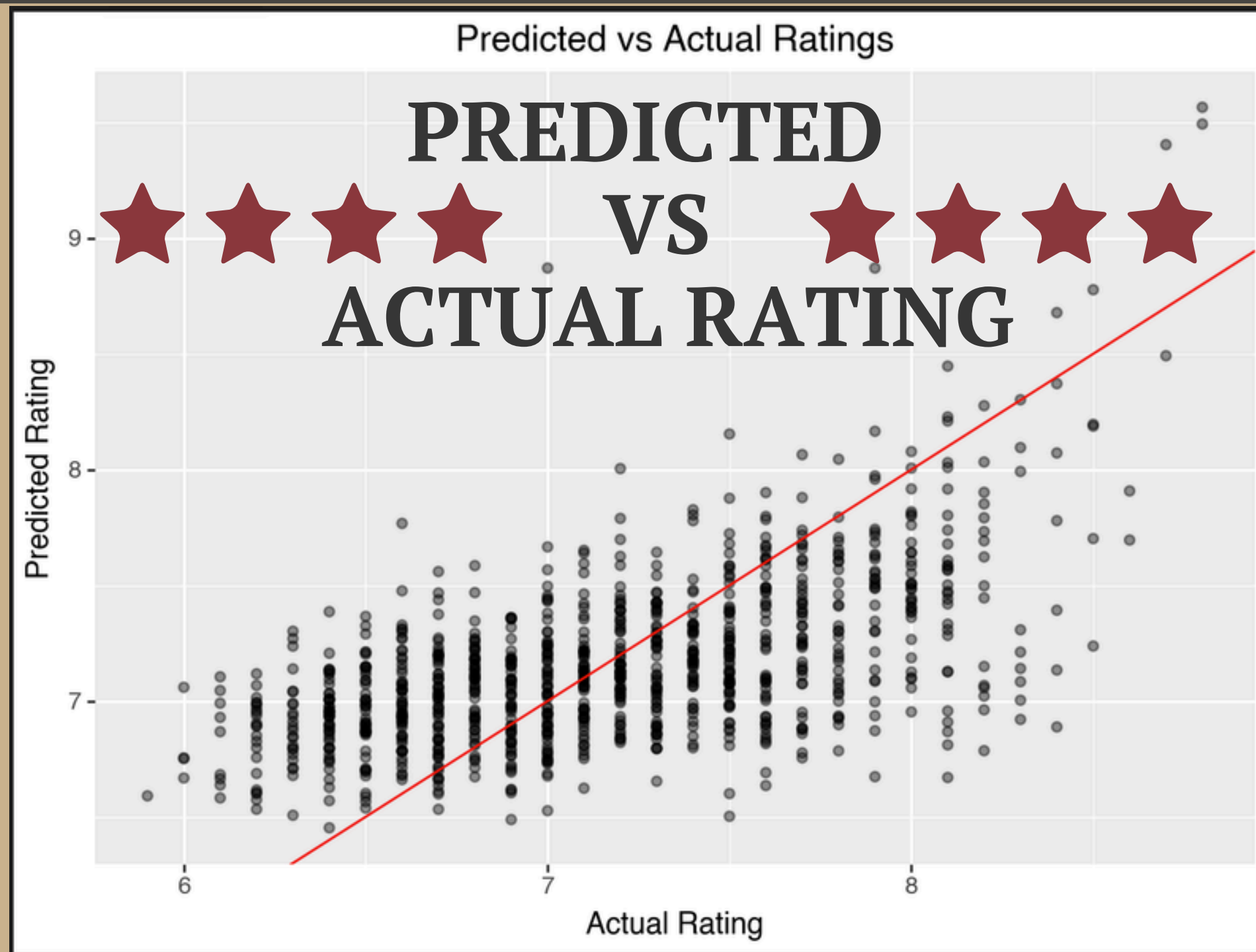
- Shows which directors appear most frequently in the dataset
- Woody Allen has the highest number of movies
- Followed by Clint Eastwood and Steven Spielberg
- Helps us understand movie distribution across directors



Top 5 Directors with the Most Movies in the Dataset



0123456789



0123456789

IMDB TOP 5000 MOVIES

- Best model: Ridge Regression ($\alpha = 0.1$)
- Test MSE: 0.21
- Prediction error(RMSE)
 ≈ 0.46 rating points
- Good accuracy given rating range (6–9.3)

	Feature Set	Model	Mean CV MSE
0	base	Linear Regression	0.346923
1	movie	Linear Regression	0.246131
2	full	Linear Regression	0.244186
3	base	Ridge Regression $\alpha=1$	0.346923
4	movie	Ridge Regression $\alpha=1$	0.246131
5	full	Ridge Regression $\alpha=1$	0.244186
6	base	Ridge Regression $\alpha=0.1$	0.346923
7	movie	Ridge Regression $\alpha=0.1$	0.246131
8	full	Ridge Regression $\alpha=0.1$	0.244186
9	base	Lasso Regression $\alpha=1$	0.347840
10	movie	Lasso Regression $\alpha=1$	0.347840
11	full	Lasso Regression $\alpha=1$	0.347840
12	base	Lasso Regression $\alpha=0.1$	0.347840
13	movie	Lasso Regression $\alpha=0.1$	0.275353
14	full	Lasso Regression $\alpha=0.1$	0.275353

▶ Play

- RATINGS RANGE FROM ABOUT 6 TO 9
- GENRE AND VOTES INFLUENCE RATINGS
- WESTERN HAS HIGHEST AVERAGE
RATING
- RIDGE REGRESSION PERFORMS BEST
- TEST MSE: 0.21 (~0.46 RATING ERROR)

SUMMARY



THANKS FOR

YOUR ATTENTION

▶ End

